LITERATURE REVIEW

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Literature Review

Publication 1

The publication confirms the chronological split, preprocessing steps, final training/test sizes and model architecture.

Field	Details
Title	Returnformer: A Graph Transformer-Based Model for Predicting Product Returns in E-Commerce
Harvard reference	Cao, Q., Zhang, N. and Li, H. (2026). 'Returnformer: A Graph Transformer-Based Model for Predicting Product Returns in E-Commerce'. *Entropy*, 28(1), p.72. Available at: https://10.3390/e28010072 [Accessed: 1 September 2026].
Link	Article Link: https://www.mdpi.com/1099-4300/28/1/72 Dataset Link: https://osf.io/c793h/overview
Modelling	The proposed Returnformer model is based on a Graph Transformer and operates on a customer–product bipartite graph. Node2Vec is used to obtain global topological embeddings, which are combined with the original node features. A Graph Transformer captures long-range customer–product dependencies, while graph-level external attention supports information propagation between subgraphs. A Kolmogorov–Arnold Network (KAN) is used as the decoder. The study compares Returnformer with MLP, XGBoost, LightGBM, CatBoost, GCN, GAT and GraphSAGE. The reported Returnformer configuration includes two Graph Transformer layers, four attention heads, an embedding dimension of 128, dropout of 0.45 and a batch size of 128.
Preprocessing techniques	The researchers checked for duplicates, missing values and outliers. Duplicate records were removed. Customer and product IDs were recoded. Customer birth year was converted into age. Approximately 30,000 customers aged over 90 were treated as outliers and excluded. Categorical variables were one-hot encoded, while numerical variables were standardised. The data was separated chronologically to reduce the risk of data leakage. A customer–product bipartite graph was then constructed.
Datasets used	A real-world clothing dataset supplied by ASOS, a UK fast-fashion e-commerce company. The data contains customer keep/return events from September–November 2021, with customer and product information. The task is binary classification: returned = 1 and kept = 0. Approximately 1.8 million events were available before preprocessing. After preprocessing, the training set contained 939,537 events and the test set contained 858,526 events.
Key takeaways	1. Return prediction can benefit from combining customer, product and historical interaction information rather than treating every transaction independently.

	2. Careful preprocessing, particularly removal of duplicates/outliers and prevention of temporal data leakage, is important for reliable return prediction. 3. Graph-based models can capture relationships between customers and products, however the approach is considerably more complex than conventional tabular machine-learning models. For this project, graph modelling provides a useful, advanced approach to consider, while simpler models can provide an appropriate baseline.

Publication 2

Field	Details
Title	Proactive Return Prediction in Online Fashion Retail Using Heterogeneous Graph Neural Networks
Harvard reference	Ma, S. and Wang, W. (2024). 'Proactive Return Prediction in Online Fashion Retail Using Heterogeneous Graph Neural Networks'. *Electronics*, 13(7), p.1398. Available at: https://10.3390/electronics13071398 [Accessed: 1 September 2026].
Link	Article Link: https://www.mdpi.com/2079-9292/13/7/1398 Dataset Link: https://github.com/Shawn-nau/HGNN
Modelling	The proposed COP-HGNN (Customer–Order–Product Heterogeneous Graph Neural Network) models relationships between customers, orders and products. The authors compare the proposed model with several traditional machine-learning models: Logistic Regression, Naïve Bayes, K-Nearest Neighbours, CART, Random Forest, Gradient Boosting Decision Trees (GBDT) and LightGBM. The heterogeneous graph represents customers, orders and products as different node types, while order-product relationships are represented through edges.
Preprocessing techniques	The preprocessing pipeline included data cleaning, data transformation, feature engineering, normalisation and data splitting. Temporal variables such as month of order and day of week were transformed into usable model features. Order-level variables such as number of products, total order value, average order value and discount rate were derived. Relevant customer, product and order attributes were incorporated into the graph structure and numerical variables were normalised. The data was divided chronologically so that historical information was used to predict later returns.
Datasets used	A real-world dataset from an international online fashion retailer containing 2,627,927 valid records covering approximately two years of order-level transactions and return records. The data includes customer, product and order information such as product ID, customer ID, order ID, quantity, price, coupons, payment method, colour, size, product group and returned quantity.
Key takeaways	1. Customer, order and product information can be combined to provide a richer representation of return behaviour. 2. Traditional models such as Random Forest, GBDT and LightGBM provide strong baselines for return prediction and should be considered before implementing more complex neural-network approaches. 3. Feature engineering is important as variables such as order size, order value, discount rate and product characteristics can capture behavioural patterns associated with returns.

Publication 3

Field	Details
Title	Evaluating Predictive Models for Reducing E-Commerce Returns and Associated Carbon Emissions
Harvard reference	Kalidindi, R., Narkedamilly, L. and Saladi, U.M. (2026). 'Evaluating Predictive Models for Reducing E-Commerce Returns and Associated Carbon Emissions'. *International Journal of Science and Research (IJSR)*, pp.1472–1481. Available at: https://10.21275/SR26717182049 [Accessed: 1 September 2026].
Link	Article Link: https://10.21275/sr26717182049. Dataset Link: https://www.kaggle.com/datasets/sayalikhot21/synthetic-dataset-for-e-commerce-return-analysis
Modelling	The publication investigates predictive modelling of e-commerce returns using machine-learning classification approaches and connects predicted return reduction to potential environmental benefits. The study focuses on using predictive modelling to identify likely returns and subsequently estimate the potential reduction in return-related carbon emissions.
Preprocessing techniques	The study uses the synthetic e-commerce return dataset containing order, product, customer, payment, shipping and return information. The data is prepared for predictive modelling by using relevant transaction and customer/product variables and treating return status as the prediction outcome. Categorical and numerical variables are suitable for transformation into machine-learning inputs, while variables that directly reveal the outcome after a return should be excluded when constructing a pre-return prediction model to avoid data leakage.
Datasets used	The study uses the Synthetic Dataset for E-Commerce Return Analysis, which contains simulated e-commerce transactions and return information. The dataset contains information relating to orders, products, users, return behaviour, prices, quantities, payment methods, shipping methods and discounts. The dataset is synthetically generated and does not represent real customer or transaction data.
Key takeaways	1. Predictive return modelling can support proactive interventions intended to reduce the number of unnecessary returns. 2. Return prediction can have a wider business impact than classification accuracy alone as reducing returns can also reduce reverse-logistics activity and associated environmental impacts. 3. For this project, the target should be defined so that only information available before the return occurs is used for prediction; variables such as return date, return reason and days to return should therefore not be used as predictive inputs.

Publicly Available Dataset to use in Part B

Field	Details
Name	Synthetic Dataset for E-Commerce Return Analysis
Harvard reference	Sayali Khot (2024). *Synthetic Dataset for E-Commerce Return Analysis*. Kaggle.com. Available at: https://www.kaggle.com/datasets/sayalikhot21/synthetic-dataset-for-e-commerce-return-analysis [Accessed 1 Sept. 2026].
Link	https://www.kaggle.com/datasets/sayalikhot21/synthetic-dataset-for-e-commerce-return-analysis
Number of instances	10,000 records. The dataset is provided as a CSV file. The Kaggle dataset description does not specify separate official training and testing splits, so the 10,000 records should be treated as the complete dataset before you create your own training/validation/test split.
Number of feature columns	17 original columns. Return_Status is the target variable. Therefore, there are 16 potential input columns and 1 target column, giving 17 columns in total.
Number of classes (if applicable)	2 classes: Returned and Not Returned. The target variable is Return_Status, making this a binary classification problem.
Target range (if applicable)	Not applicable, as this is a classification problem rather than regression. The target consists of the categorical values Returned and Not Returned.
Origin	The dataset is synthetically generated for e-commerce return analysis. It does not represent real-world customer or transaction data. The creator states that it was generated using a random process and designed to contain realistic-looking information relating to orders, products, users and returns. It is intended for predictive modelling, exploratory analysis, benchmarking and feature-engineering experimentation.